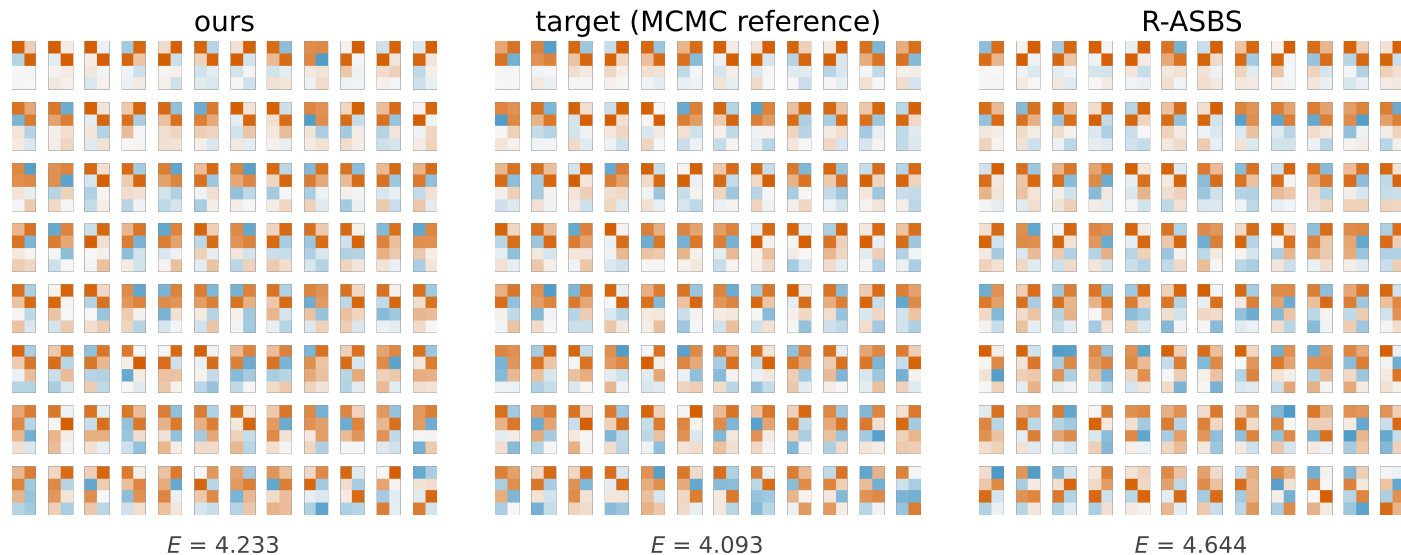


Figure 7 – $\text{St}(4,2)$ at $\beta = 2$: raw frames, and the second moment that separates the samplers

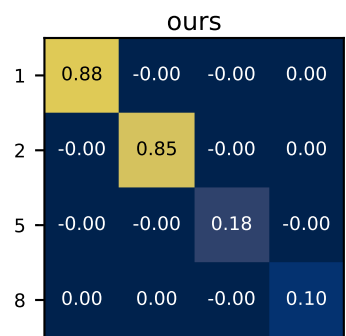
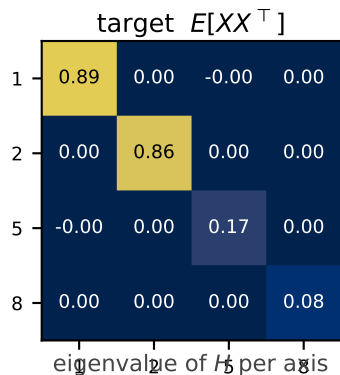


96 frames $X \in \text{St}(4,2)$, $\beta = 2$

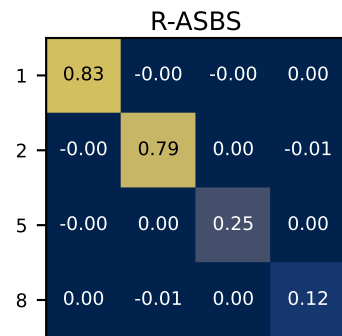
Each tile is one 4×2 matrix, drawn in the eigenbasis of $H = \text{diag}(1, 2, 5, 8)$.

Column signs are canonicalised and the frames are quantile-spaced in energy, so the three sheets line up rank for rank.

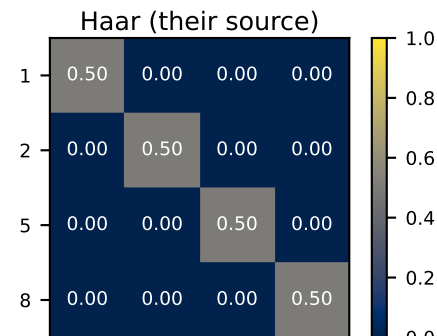
The target puts its mass in the two low-eigenvalue directions: strong colour in the top two rows of each tile, pale in the bottom two. Ours reproduces that. R-ASBS is visibly paler on top and darker on the bottom – mass has leaked back toward the uniform (Haar) frame.



$\| \cdot - \text{target} \|_F = 0.028$
4% of the way back to Haar



$\| \cdot - \text{target} \|_F = 0.138$
17% of the way back to Haar



$\| \cdot - \text{target} \|_F = 0.755$
the null: no transport at all

